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Isolation of Xanthomonadin pigment from the infected rice isolates of the bacterium *Xanthomonas campestris* pv. *oryzae*

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Abstract : The selective media is essential requirement for formation of Xanthomonadin[1] pigment. It is one of the most important characteristic feature of the genus *Xanthomonas campestris* pv. *oryzae*[2].This bacterial strain is responsible for causing leaf blight disease in rice plant. The colonies in the media containing plates were very pale yellow with colourless slime with no diffusion of pigment has taken place. These were 1 mm diameter in 2-3 days of incubation. The pathogen of three rice isolates i.e. KRH-2, Swarna & Jurly obtained from different rural areas of west singhbhum district of Jharkhand

Key words :Xanthomonadin, Pigment, Pale Yellow.

INTRODUCTION

Xanthomonas campestris pv. *oryzae* a strain of the causative agent of bacterial leaf blight disease in rice plants isolated from different rural areas of west singhbhum district of Jharkhand. Xanthomonadins are yellow, membrane-bound pigments produced by the genus *Xanthomonas campestris* pv. *oryzae*[2].

MATERIAL & METHOD

The name and method was that of Starr et. al employed for assessment of the pigment production[1]. For the pigment production following ingredients are taken i.e. Peptone-5g,Beef Extract-3g,NaCl-5g,Dextrose-10g, Agar-15g, Distilled Water-1Lit, Methanol.-3 ml. For detecting the Xanthomonadin pigments which appear to be unique to the genus is proved in practice, this would be a most useful and rapid adjunct to colony morphology for presumptive diagnosis. For the production of Xanthomonadin pigments at first media was prepared by the above ingredients. This media was then autoclaved and inoculated the pathogen to all three rice isolates i.e KRH-2, Jurly & Swarna and authentic culture (It was the freeze dried isolate of *Xanthomonas campestris* pv. *oryzae* obtained from PDDCC of DSIR, Auckland, NewZealand) separately on petridishes and control (none inoculated) was also taken separately.

After 7-14 days of incubation pale yellow coloured colonies were observed on the media. If pale yellow colony was developed then it indicate the formation of Xanthomonadin pigments. Now the medium was kept on the water bath at 90°C and the pigment was extracted as Xanthomonadin form. If the yellow coloured colony was not found, then there was not production of any Xanthomonadin pigment.

RESULTS & DISCUSSIONS

The surface colonies in the plates were very pale yellow with colourless slime. No diffusion of pigment has taken place. After incubation of 3 days, the culture become smooth, yellow, domed, undulate, mucoid, viscid, circular and convex, rather slow growing colonies (most are 1 mm diameter in 2-3 days).Colonies were pale yellow and

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colourless slime from pigment "Xanthomonadin". Production of pale yellow colony with colourless slime in the medium was true for three rice isolates and an authentic culture .This showed that three isolates of rice KRH-2, Swarna and Jurly as well as an authentic culture produce pale yellow coloured Xanthomonadin pigment and it is the characteristic feature of the bacterium *X.c.pv.oryzae*. It was observed after 7 to 14 days of incubation. Control (none inoculated) was also taken. Result is presented in fig 1.

The bacterium was gram negative, rod shaped with a single polar flagellation with copious production of yellow pigment Xanthomonadin[3]. The colonies of *X. c. pv. oryzae* are pale yellow and colourless slime from pigment "Xanthomonadin"[1]. It is most significant test for *Xanthomonas* species.The culture colour of *P. marginalis* is cream to yellowish[4]. Xanthan gum is a microbial polysaccharide obtained from *Xanthomonas campestris* pv. *oryzae*. It is of great commercial significance.[5]

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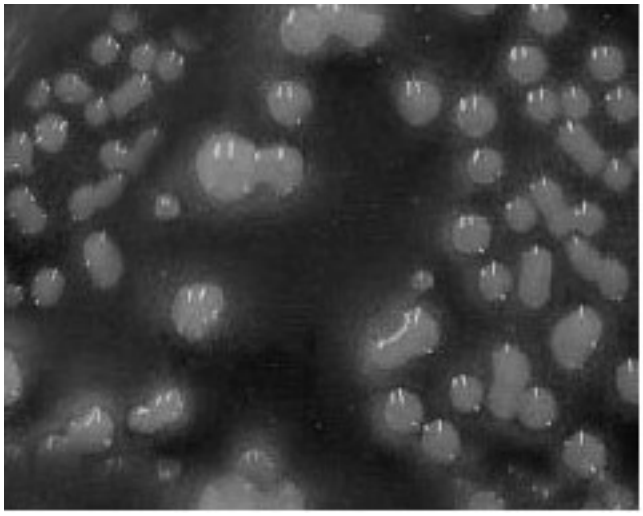


Fig.1:Xanthomonadin pigment

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